50 คน $\bar{x} = 40$ $s = 5$

① 95 %

$\alpha = 0.05$, $\frac{\alpha}{2} = 0.025$

$$Z_{0.025} = 1 - 0.025$$
$$= \phi(0.975)$$
$$= 1.96$$

0.708

$$\bar{x} \pm Z_{\alpha/2} \times \frac{s}{\sqrt{n}} = 40 \pm 1.96 \times \frac{5}{\sqrt{50}}$$

$$= (38.6123 , 41.3877) \ \#$$

② 99 %

$\alpha = 0.01$, $\alpha/2 = 0.005$

$$Z_{0.005} = 1 - 0.005$$
$$= \phi(0.995)$$
$$= 2.58$$

1.8267

$$40 \pm 2.58 \times \frac{5}{\sqrt{50}} = (38.1733 , 41.8267) \ \#$$

③ 72 %

$\alpha = 0.28$ $\alpha/2 = 0.14$

$$Z_{0.14} = 1 - 0.14 = \phi(0.86)$$
$$= 1.08$$

$$40 \pm 1.08 \times \frac{5}{\sqrt{50}} = (39.2354 , 40.7647) \ \#$$

① คา a $\rightarrow$ $1 - \ldots \%$

② คา $a/2$ $\rightarrow$ $/2$

③ $1 - (a/2) = \phi(\ldots) = \ldots$

④ $\bar{x} \pm \_\_ \times (std / \sqrt{sample})$